

INDIAN STATISTICAL INSTITUTE

Analysis on Lie groups

Mid Semester Examination: 2025-26

M.Math 2nd Year

Date: 27.02.2026

Maximum Marks: 40

Maximum Time: 2.5 hrs.

Answer as many questions as you can but maximum you can score is 40.

✓(1) Let G be a connected topological group, $U \subset G$ a non-empty open neighborhood of the identity and $U^n := \{g_1 \cdot g_2 \cdots g_n : g_1, \dots, g_n \in U\}$ for any $n \in \mathbb{N}$. Show that $G = \bigcup_{n=1}^{\infty} U^n$. [8] •

✓(2) Let H be a compact subgroup of $GL(n, \mathbb{R})$ such that $O(n) \subset H$. Show that $H = O(n)$. [10] •

(3) (a) State Schur's lemma for finite dimensional representations. [2] •

✓(b) Is the natural representation of $SO(2)$ in $\mathbb{R}^2$ irreducible? [2] •

(c) Use Schur's lemma to find the center of the group $GL(n, \mathbb{C})$. [6]

✓(4) Fix $q > 1$, and consider the group G consisting of the matrices of the form

$$\begin{pmatrix} q^k & x \\ 0 & 1 \end{pmatrix} \quad (k \in \mathbb{Z}, x \in \mathbb{R}).$$

(a) Show that G is closed subgroup of $GL(2, \mathbb{R})$. [3] •

(b) Prove that the linear form

$$f \mapsto \sum_{k \in \mathbb{Z}} \frac{1}{q^k} \int_{-\infty}^{\infty} f(k, x) dx \quad (f \in C_c(G))$$

defines a left Haar measure and that the linear form

$$f \mapsto \sum_{k \in \mathbb{Z}} \int_{-\infty}^{\infty} f(k, x) dx \quad (f \in C_c(G))$$

defines a right Haar measure. [8] •

(c) Determine the modular function of G . [3]

[P.T.O]

✓(5) Let $G = SU(2)$, and $T := \{t_\phi : \phi \in \mathbb{R}\}$ where

$$t_\phi := \begin{pmatrix} e^{i\phi} & 0 \\ 0 & e^{-i\phi} \end{pmatrix} \quad (\phi \in \mathbb{R}).$$

(a) Show that T is a commutative compact subgroup of G . Show that every element of G that commutes with T belongs to T . [3+3]

(b) Let χ_n be the character of the irreducible representation π_n (as defined in the class!) of G for $n \in \mathbb{N}$. Show that

$$\chi_n(t_\phi) = \frac{\sin(n+1)\phi}{\sin \phi}.$$

[6]

(c) Prove the following **Clebsch-Gordon formula**, for $m, n \in \mathbb{N}$ with $m \geq n$:

$$\pi_m \otimes \pi_n \sim \pi_{m+n} \oplus \pi_{m+n-2} \oplus \cdots \oplus \pi_{m-n}.$$

[6]

(Here π_n denotes the representation of $SU(2)$ in $P_n(\mathbb{C}^n)$ (consisting of homogeneous polynomials in $\mathbb{C}^2$ of degree n) induced from the natural action of $SU(2)$ on $\mathbb{C}^2$)

(6) (a) State the Schur orthogonality relations for matrix coefficients. [2]

✓(b) Find the left Haar measure of $GL(n, \mathbb{R})$. [3]

(c) Let χ be the character of the natural representation of $SO(n)$ in $\mathbb{C}^n$. Let $d\mu$ stand for the normalised Haar measure on $SO(n)$. Determine the integral

$$\int_{SO(n)} |\chi(g)|^2 d\mu(g).$$

[5]